



**School of
Professional Studies**

DATA 607

Data Acquisition and Management

Prerequisites	None
Semester	Spring 2026

Course Relevance

In professional data analytics and data science roles, the majority of effort is spent on **acquiring, cleaning, integrating, and validating data** rather than on modeling itself. This course develops those core capabilities.

Recent advances in machine learning frameworks and generative AI have made model development faster and more automated. As a result, competitive advantage has shifted toward **data-centric AI**—the ability to obtain high-quality, well-structured, and well-governed data for downstream analytics and machine learning. In this environment, the skills taught in this course are among the most valuable in the data science profession.

Course Description

This course is designed around a single guiding question: **What do employers expect from entry-level data scientists?**

Accordingly, most of your weekly effort will focus on **designing, coding, and explaining data pipelines and analyses** that resemble real workplace deliverables. Employers expect graduates to combine **strong technical foundations, high-quality work products**, and the ability to **use generative AI effectively to increase coding productivity**.

A core challenge—both in this course and in professional practice—is to avoid using AI tools in ways that weaken your underlying understanding. The course therefore emphasizes **sound scientific practice, reproducible workflows, and professional code patterns** so that your outputs remain both efficient and technically rigorous.

Throughout the semester, you will work in an environment that prioritizes **iteration, experimentation, and collaboration**, mirroring modern data science teams.

The course focuses on **structured (rectangular) and semi-structured data** such as **JSON, HTML, and text**. The primary technical tools are **R and SQL**.

Grading Distribution

Component	Weight
Assignments (6 x 50)	30%
Projects (3 x 90)	27%
Final Project Proposal (1 x 20)	2%
Final Project (1 x 150)	15%
Final Project Presentation (1 x 30)	3%
Discussion and Reading Quizzes (14 x 10)	14%
Data Science in Context Presentation (1 x 50)	5%
TidyVerse recipes	4%
TOTAL	100%

Late work is not accepted for grading. However, there are eight assignments and four projects assigned, and your final grade is based on your six highest-scoring assignments and your three highest-scoring projects.

Assignment Availability

Each course week will be available on the previous Thursday at 6:00 a.m. ET.

All information on the four course projects, the Data Science in Context presentation, the two TidyVerse assignments, and the three Final Project deliverables can be found by selecting Content on the Brightspace top level menu, then scrolling down to Major Projects on the right hand menu.

Grading Scale

Quality of Performance	Letter Grade	Range %
Excellent - work is of exceptional quality	A	93 - 100
	A-	90 - 92.9
Good - work is above average	B+	87 - 89.9
Satisfactory	B	83 - 86.9
Below Average	B-	80 - 82.9
Poor	C+	77 - 79.9
	C	70 - 76.9
Failure	F	< 70

Course Completion Requirements

To pass this course, you must complete:

- At least six assignments
- Three projects
- The final project
- Final project presentation

If you cannot deliver your final presentation in our final Meetup, you'll need to make available a recorded version of your final presentation before our final meetup.

Hands-On Labs

There are some short ungraded hands-on labs that will help you prepare for your weekly programming assignments and projects. You don't need to turn these in.

Some Discussion Assignments include a short quiz on an associated reading or viewing assignment as part of the overall grade.

Discussions, Presentations, and TidyVerse Recipes

While this material is important, please note that this work makes up 23% of your grade. Please do the readings and participate in the discussions and any discussion-related group assignments, make your Data Science in Context presentations, and participate in the creation and editing of TidyVerse recipes on the shared GitHub site.

If you have limited time for the course, please remember to invest most of your efforts in completing the projects and assignments. The assignments merit close attention because they will help you to be successful on the projects.

Data Science in Context presentations are open to a topic of your choosing related to the course material and data science in general. They are 5 minutes only and will take place as we dive into the week's material in the weekly meetups. Up to three presentations will take place on each Meetup call. Make sure to schedule your presentation early by using the respective Forum in the Discussion Forum.

Reproducibility and Testing

Students are responsible for providing all code and data so that I can test your work. If you turn in code that does not run, you will not receive credit, unless you also include an explanatory note at the time of submission.

At the same time, you don't need to turn in perfect code. Generous partial credit will be given for deliverables that are timely, tested, and reproducible. Cutting corners—as long as they are documented at the time of submission—is also acceptable.

Groupwork

Groupwork is encouraged on most projects and assignments and required on Project 3. Effective virtual collaboration is highly valued in the data science marketplace; because of its interdisciplinary nature, much of the work that needs to be done requires more than one person, and increasingly often at multiple locations.

Earning a Grade of A

If you complete the course work correctly and on time, you'll comfortably pass the course. A grades will be reserved for students that go above and beyond, such as consistently taking on challenge assignments.

Course Generative AI Policy

Overview

This course permits and encourages the use of large language models (LLMs) and other generative AI tools in completing assignments. You may use these tools in any way you see fit—for brainstorming, code generation, debugging, explanation, drafting, editing, or any other purpose that advances your work.

Requirements for Use

When using generative AI tools, you must fulfill two requirements:

1. Provide a Transcript

Submit a complete transcript of your interactions with any LLM used in completing an assignment. This transcript must include all prompts you provided and the responses you received. Most platforms allow you to export or copy conversation histories; screenshots are also acceptable.

2. Cite the Models Used

Include a citation for each AI tool you used. Citations should specify the model name, version (if known), the developer, and the date of access.

Example Citations

- OpenAI. (2025). ChatGPT (Version 5.2) [Large language model]. <https://chat.openai.com>. Accessed January 8, 2026.
- Google DeepMind. (2025). Gemini 3 Flash [Large language model]. <https://gemini.google.com>. Accessed January 8, 2026.
- Anthropic. (2025). Claude Opus 4.5 [Large language model]. <https://claude.ai>. Accessed January 8, 2026.

Free Access to AI Tools

All students have free access to Google Gemini through their university accounts. You are encouraged to take advantage of this resource throughout the course. Here is the link: <https://search.app/RdW9w>

A Note on LLMs, Learning, and Professional Development

This course has two goals: **to build foundational knowledge** in data science methods and concepts, and to help you **deliver excellent work product**. Your grade is determined solely by the quality of the work you submit.

However, the way you choose to interact with LLMs will have a profound effect on your ability to build and extend the foundational knowledge you need for long-term success in this field.

Consider the difference between asking an LLM to "write code that does X" versus asking it to "explain the intuition behind algorithm Y, then walk me through how I might implement it step by step." Both approaches may produce a working solution, but only one builds your capacity to solve novel problems independently, to recognize when an AI-generated answer is subtly wrong, and to push the boundaries of what's possible in your future work.

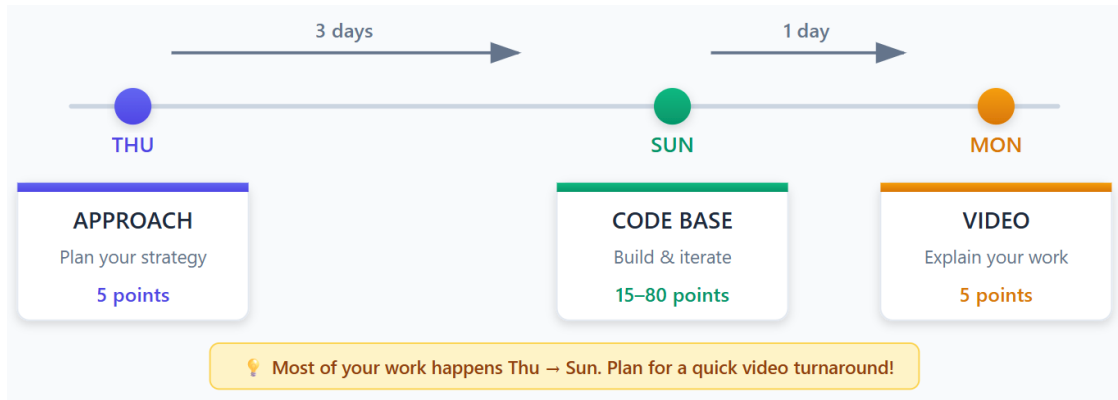
You are training to be experts. Experts understand why things work, not just that they work. Use these tools in ways that serve your development as a practitioner—not just your grade on any single assignment.

The transcripts you submit are not graded for "correct" usage. They are required so that you develop the habit of reflection and transparency, and so that you can look back on your own learning process over time.

General Requirements for Assignments and Projects

Overview

Most assignments have three sequential deliverables designed to encourage iterative engagement, mirror real-world analytics workflows, and support early feedback.



Deliverable	Due	Points
Approach	Thursday end of day ET	5
Code Base	Sunday end of day ET	15–80 (varies)
Video Explainer	Monday end of day ET	5
No work is accepted for grading past its deadline.		

Submission Requirements

For each deliverable, submit:

- A link to your GitHub repository
- Either: **A link to your .Qmd/.Rmd published on RPubs**, or a link to an HTML file in your GitHub repository

Your repository must be public or have acatlin added as a collaborator.

Approach Deliverable

Create a starter .Qmd or .Rmd file with an introductory section describing your planned approach. Go beyond restating the requirements:

- How will you tackle the problem?
- What data challenges do you anticipate?

No code is required at this stage.

Code Base Deliverable

Continue working in the same file from your Approach deliverable. Your commit history should show evidence of iteration—it's expected that your approach may evolve as you code.

Structure Requirements

Section	Requirements
Introduction	2–3 sentences describing your chosen article, with a link
Body	Your analysis (additional graphics/tables welcome but not required). Provide comments in text sections that document your workflow.
Conclusions	Findings and recommendations; ideas to extend, verify, or update the work

Each text section must include at least one header plus additional narrative text.

Reproducibility

Your code must run without errors in my environment. Ensure all data files are accessible via URL (e.g., GitHub, AWS S3)—not local file paths.

Tips

- The best analyses start with a business question, translate it into a data question, and provide a data-driven answer.
- Generous partial credit is given, but non-submissions receive zero grades.

Video Explainer Deliverable

Record a 3–7 minute video (target: ~5 minutes) explaining your work.

Suggested Topics

- How and why you selected your dataset
- Whether and how you used an LLM
- Notable or challenging code
- Key insights from your analysis
- Potential next steps or extensions

Guidelines

- Show visuals or code selectively rather than narrating slides
- Face-on-camera is optional
- Production quality is not graded
- **You must understand all code you submit.** Be prepared to explain any LLM-generated code.

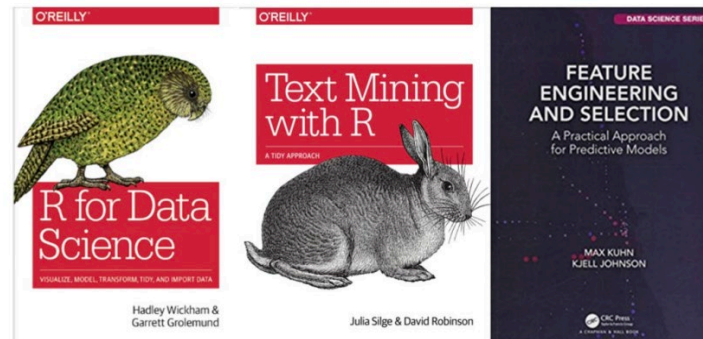
Submission

Upload your video file (e.g., MP4) or provide a URL link.

Free recording tools: [Loom](#) | [ScreenPal](#)

Course Learning Materials

Required Texts



1. **R for Data Science (2e)** by Hadley Wickham and Garrett Grolemund. This is the primary text for the course. Freely readable at: <https://r4ds.hadley.nz/>
2. **Text Mining with R: A Tidy Approach** by Julia Silge and David Robinson. O'Reilly, 2017. Freely readable at: <https://www.tidytextmining.com/>
3. **Feature Engineering and Selection: A Practical Approach for Predictive Models** by Max Kuhn and Kjell Johnson. Chapman & Hall/CRC, 2019. Freely readable at: <http://www.feats.engineering/>

Print copies for each of these texts are available for purchase on Amazon and elsewhere.

Recommended Texts

- **The Language of SQL, 3rd Edition** by Larry Rockoff. ISBN: 978-0137632695
- **Practical SQL, 2nd Edition** by Anthony DeBarros

Alternatively, there are many excellent online resources, such as <http://sqlzoo.net>

Optional Recommended Text

Data Science for Business by Provost, F., & Fawcett, T. (2013). Sebastopol, CA: O'Reilly. This textbook provides good insights into the Data Science discipline in the business world. Consider using this textbook *only if available to you* for the weekly discussions and general reference. It is not a coding book, but can help germinate ideas for some of the projects, including your final project.

Software and Tools

We will make use of the R programming environment and the RStudio IDE. We will use other open source software, including PostgreSQL and MongoDB. Details for obtaining and installing the appropriate software will be provided in the course materials. All of the software will work on (or from) both PCs and Macs.

Instructor Information

Instructor	Andy Catlin
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Phone	(616) 638-8344

How This Course Works

Weekly Meetups

Meetups take place every week on Wednesdays from 6:45 p.m. to 7:45 p.m. ET.

Zoom Link: <https://us06web.zoom.us/j/99498039934>

You are strongly encouraged to attend; all meetups will be recorded. You are not required to attend the meetups, but you are responsible for watching the recording and reviewing any meetup collateral if you're not able to attend. Meetup recordings can be found in Brightspace by selecting Content from the top level menu, then scrolling down to the bottom of the right hand menu.

Office Hours

- **Occasional Weekend Office Hours:** A few times during the semester, we'll have optional additional office hours on topics of interest, especially around data engineering.
- **Regular Office Hours:** Can be scheduled by email appointment. If you need extra help and are willing to invest the time and effort to be successful, I'll make the time to help you. However, you should not be asking for extra help on a project the day before it's due, since this indicates that you're not investing the time and effort to be successful.

Communication

You are encouraged to ask questions that might be of interest to other students on the course Slack Channel.

For the most part, you can expect me to respond to questions by email or Slack within one business day.

Course Schedule

Course deliverable dates are available in Brightspace under **Tools** → **Calendar**. We will also review upcoming deliverables at the start of each weekly meetup.

Unit	Topic	Core Readings
Wk. 1	Data Ethics; R: Data Types and Basic Operations	R for Data Science (2e), chapters 1, 2, 3, 7, 14, 15, 27, 28
Wk. 2	R and SQL	R for Data Science (2e), chapters 8, 9, 10, 23
Wk. 3	R: Character Manipulation and Date Processing	R for Data Science (2e), chapters 14, 15, 16, 17, 18, 19
Wk. 4	R: Exploratory Data Analysis; Data Imputation	R for Data Science (2e), chapters 11, 12, 13, 20
Wk. 5	R: Working with Tidy Data	R for Data Science (2e), chapters 4, 5, 6
Wk. 6	R: Data Transformations; Feature Engineering	Feature Engineering and Selection, chapter 1
Wk. 7	Web Technologies	R for Data Science (2e), chapter 25
Wk. 8	Scraping Web Pages	R for Data Science (2e), chapter 26
Wk. 9	Working with Web APIs	httr2 quickstart vignette
	Spring Break	—
	Spring Break	—
Wk. 10	Natural Language Processing	Text Mining w/ R, chapters 1–4
Wk. 11	Recommender Systems	Mining Massive Datasets, chapter 9
Wk. 12	Graph Databases	Selected coursework from Neo4j GraphAcademy
Wk. 13	Generative AI	No Readings
Wk. 14	Final Presentations	No Readings

University Policies

Accessibility and Accommodations

The CUNY School of Professional Studies is firmly committed to making higher education accessible to students with disabilities by removing architectural barriers and providing programs and support services necessary for them to benefit from the instruction and resources of the University. Early planning is essential for many of the resources and accommodations provided.

More information: http://sps.cuny.edu/student_services/disabilityservices.html

Online Etiquette and Anti-Harassment Policy

The University strictly prohibits the use of University online resources or facilities, including Brightspace, for the purpose of harassment of any individual or for the posting of any material that is scandalous, libelous, offensive or otherwise against the University's policies.

More information: [CUNY Online Etiquette Policy](#)

Academic Integrity

Academic dishonesty is unacceptable and will not be tolerated. Cheating, forgery, plagiarism and collusion in dishonest acts undermine the educational mission of the City University of New York and the students' personal and intellectual growth.

More information: [CUNY Academic Integrity Policy](#)

Student Support Services

If you need any additional help, please visit Student Support Services:

http://sps.cuny.edu/student_resources/